MoralityBench: An exploratory benchmark of elicited moral profiles in generative and structured decision models

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Working manuscript, updated October 29, 2026

Abstract

MoralityBench describes the moral response profiles elicited from AI systems using an adapted 36-item Moral Foundations Questionnaire 2 (MFQ-2) and the 20-item Ethics Position Questionnaire (EPQ). This updated manuscript reports 20 scored generative model configurations, each with five questionnaire administrations, plus Jev, a separately evaluated structured decision system. The combined generative archive contains 5,415 usable ratings from 5,600 scheduled items. The original cohort comprises 13 configurations, with 3,580 usable ratings from 3,640 scheduled items; the October 14 extension adds GPT-6 Astra, Muse Wonder, and Gemini 4 Argon, with 715 usable ratings from 840 scheduled items. The October 29 extension adds GPT-6.1 Terra, Claude Haiku 5.5, Kimi K3.1 Pro, and MiniMax M3.1 Flash, with all 1,120 responses usable. In the original cohort, excluding missing ratings, exact agreement across all ten pairs of runs ranges from 60.4% to 94.7%. Within the original cohort, DeepSeek V4.1 Flash has the lowest average distance from the six published US foundation means (0.239), followed by MiMo v2.6 Pro (0.253); MiMo has the lowest distance in three of the four new runs. Five configurations cross an EPQ category boundary between runs. In the original snapshot every configuration exceeded the human Purity reference, but DeepSeek's five-run mean falls below it, and the replacement of four of the six Purity items limits what that comparison can show. Jev returns all 280 ratings, with 95.7% agreement after rounding. Its prompt and structured interface differ from the generative protocol, so these observations cannot establish an architectural advantage. The release includes individual responses, completion records, scoring code, and a public leaderboard of five-run averages and agreement percentages. The October 14 and October 29 models use strict whole-answer integer scoring; the older models were not retested and retain their original scoring. Because many of Gemini's responses are verbose or missing, its profile depends on the parser, and rejecting digits taken from its prose reverses its distance ordering relative to Astra. These descriptive results concern responses to adapted questionnaires and do not establish equivalent human traits, moral agency, ethical superiority, or deployment safety.

Keywords: AI evaluation; moral psychology; moral foundations; ethical ideology; survey validity; structured decision models.

1 Introduction

AI systems give advice about welfare, fairness, obligations, and conflicts between social commitments. Evaluating that advice requires knowing which commitments a response expresses and how the question elicited them. Calling a system "aligned" leaves several questions unanswered. Does its answer resemble a population average? Does it meet a stated ethical standard? Would different wording produce the same answer, and does the answer predict behavior outside the questionnaire?

Existing benchmarks examine different parts of this problem. ETHICS tests predictions about ethical judgments [1]. MoralChoice examines choices in moral scenarios, including uncertainty and consistency [2]. MoralBench adapts moral-foundations questionnaires and vignettes for model evaluation [3]. Because each covers only part of the question, each needs a clear account of what it measures and how it administers and scores the task.

MoralityBench combines two moral-psychology instruments in a compact resource whose responses and scores are open to inspection. The adapted MFQ-2 distinguishes equality from proportionality: a respondent can endorse equal treatment or outcomes while taking a different position on allocating rewards according to contribution. The EPQ measures idealism and relativism, which describe ethical positions rather than the content of a foundation profile. Together, the two instruments expose differences that an overall score can hide.

We use *profile* to mean the scores produced by a specified interface, prompt, item set, and scoring rule. Those scores do not establish that a model has a stable personality, private convictions, or moral agency. A questionnaire validated for human respondents needs further validation when researchers adapt it for machines, and biases in how models answer surveys add another source of uncertainty [7].

After describing the instruments and administration procedures, we recompute the published foundation and ethical-position scores from the individual ratings, check completion rates, and examine how missing items, the Purity adaptation, category thresholds, and Jev's score conversion affect the results. Throughout, we separate findings in the released data from explanations that would require another experiment.

2 Related work and measurement scope

2.1 Moral evaluation benchmarks

ETHICS covers commonsense morality, justice, deontology, virtue, and utilitarianism, and tests whether a system predicts specified judgments in those domains [1]. MoralityBench instead records endorsement profiles; its questionnaire items have no designated morally correct answer.

MoralChoice asks models to choose between actions in scenarios that differ in moral ambiguity. Because it examines uncertainty and sensitivity to wording [2], the way a question is asked becomes part of the evaluation. The MoralityBench snapshot contains no equivalent experiments with repeated wording or action choices, so it cannot establish comparable response stability.

MoralBench uses adapted MFQ-30 items and moral-foundations vignettes for binary and comparative assessments [3] and is a separate project from MoralityBench. Moral-foundations benchmarks and evaluations of moral responses in language models predate this work. Our release combines an adapted MFQ-2 with EPQ scores, item completion records, and a separate structured decision comparator.

2.2 Human instruments and machine administration

The MFQ-2 measures Care, Equality, Proportionality, Loyalty, Authority, and Purity [4], and Forsyth's EPQ measures idealism and relativism [5]. We use these dimensions to describe model answers; their use here does not amount to a validated psychological assessment of an AI system.

Ordering and labeling effects in language-model surveys show why the response process needs

scrutiny [7]. An apparently plausible answer may reflect a learned convention, compliance with the prompt, or a preference for one answer format. We therefore restrict our empirical claims to the recorded ratings. The data alone cannot identify effects of training, alignment methods, or architecture.

3 Benchmark design

3.1 Resource and unit of analysis

The public release contains instrument definitions, administration scripts, individual ratings, scored summaries, and a website for comparing profiles [9]. The instrument metadata identifies version 1.0, created October 2, 2026, and Appendix A identifies the snapshot used here. We analyze each *recorded model configuration* under its repository label and requested API identifier, which we retain so others can attempt replication. The stored ratings do not authenticate the backend checkpoint that answered a request.

The original generative scripts name 14 configurations, of which 13 appear in the original scored release. Muse Spark 1.3 has 56 missing ratings, so it has no score to compare, and Jev is reported separately. The initial audit uses the released records and published human reference means, and Section 6 reports four additional administrations of each scored LLM. The October 14 extension adds three scored configurations, bringing the manuscript and dated leaderboard to 16 scored generative configurations plus Jev. The October 29 extension subsequently adds four scored configurations, bringing the current manuscript and leaderboard to 20 plus Jev. The three dated cohorts retain their respective collection dates and parsing rules. No part of this study involved new human participants.

3.2 Adapted MFQ-2

The foundation questionnaire has 36 items, six for each foundation, scored as integers from 1 to 5. The release combines Care and Equality into an individualizing score and Proportionality, Loyalty, Authority, and Purity into a binding score. We report the six foundations as the main result because composite scores can hide differences between them.

The released questionnaire changes the original wording. Its metadata lists substitutions such as people to entities and country to collective, and it identifies items 6, 12, 30, and 36 as custom replacements. All four are Purity items, so replacements account for two-thirds of that subscale. The changes are meant to make statements about people applicable to nonhuman respondents, but they may also change what the statements measure. We call this version the *adapted MFQ-2* and treat comparisons with human scores on the original questionnaire as exploratory.

For configuration m and foundation f, the observed-item mean is

$$s_{mf} = \frac{1}{n_{mf}} \sum_{i \in A_{mf}} r_{mi}, \tag{1}$$

where A_{mf} is the set of items with available numeric ratings and $n_{mf} = |A_{mf}|$. Each foundation has six scheduled items. Missing values are omitted rather than replaced with a neutral answer, and an empty subscale is undefined. We report n_{mf} alongside s_{mf} because a score based on three answered items is not directly equivalent to one based on six.

3.3 EPQ and operational categories

The EPQ component contains 20 items scored from 1 to 9. Idealism is the observed mean of items 1 to 10; relativism is the observed mean of items 11 to 20. Continuous coordinates are the primary result. To reproduce the current display, we classify a dimension as high when its mean is at least 5. High idealism and low relativism yield Absolutist; high values on both yield Situationist; low idealism and high relativism yield Subjectivist; low values on both yield Exceptionist.

This fixed *scale-midpoint rule* assigns a score of 5 to the high category. The instrument metadata describes a median split, but the displayed categories use the scale midpoint, and the cutoff has not been validated against a human reference population. It is a display convention: a small change near 5 can change a category even when the scores remain similar.

3.4 Human reference and profile distance

The repository uses the US reference vector

$$\mathbf{h} = (4.05,\ 2.88,\ 3.63,\ 2.81,\ 3.01,\ 2.26) \tag{2}$$

in the order Care, Equality, Proportionality, Loyalty, Authority, and Purity. These values match Table 3 of Zakharin and Bates [6]. The publication is dated 2026, whereas the benchmark metadata cites 2025, and the article reports different US sample sizes in its abstract (835) and Methods (830). We use its published table means without claiming to resolve that discrepancy or treating the sample as an exhaustive description of the US population.

The displayed overall distance is an equally weighted mean absolute difference:

$$D_m = \frac{1}{6}\sum_{f=1}^{6} |s_{mf} - h_f|. \tag{3}$$

Distance is measured in Likert points, calculated from unrounded item means, and rounded for display. The calculation does not account for uncertainty in the human reference, variation between people, covariance, or differences in reliability across constructs. A smaller value means only that the six means are closer to the chosen reference. It establishes neither ethical quality nor safety, and it says little about how well a model represents any particular person.

4 Administration and audit

4.1 Generative configurations

The released scripts send one item per OpenRouter chat-completion request, using a single user message and no explicit system message. They process the instruments in stored order; although the metadata mentions randomization, the scripts do not randomize the items. Each request sets temperature to 0, allows up to 4096 output tokens, and includes `reasoning: {exclude: true}`. That setting excludes returned reasoning but does not show that the backend performs no internal reasoning.

The prompt asks for an immediate numeric answer without explanation and tells the model not to object that the questions do not apply to LLMs. This instruction encourages the model to accept

the questionnaire's frame, but it cannot establish that a first-person model answer has the same meaning as a human self-report.

The parser accepts the first digit in the returned text without checking that the full response is a single integer on the permitted scale. If a request raises an exception, the runner tries up to three times before recording a missing value, and a response with no digit is also recorded as missing. The generative result files retain item identifiers and parsed ratings but discard response text, provider metadata, and exception details, so they cannot distinguish a refusal from an empty response, a parsing failure, or a transport error. We therefore call these entries *missing*, although the scored file labels them refusals.

4.2 Jev as a separate comparator

TypeSafe describes Jev as a System One model that returns structured decision values instead of generating answer strings one token at a time [8]. Following that documented interface distinction, we call Jev a non-generative structured decision comparator. We have not audited its internal architecture or tested the vendor's claims about calibrated uncertainty.

The Jev runner calls `jev-latest` through the System One API, batching each instrument into one request. Each item uses the Score primitive, with the Likert anchors as criteria. The request state asks how strongly statements accord with ethical principles and moral reasoning, and item instructions ask how well a statement describes a moral position or ethical belief. The generative prompt instead asks how well the statement describes the respondent or whether the respondent agrees. Task framing therefore changes along with interface, batching, and decoding.

The runner converts Jev's zero-indexed score x to $\mathrm{round}(x) + 1$ and clips it to the instrument's range, using Python rounding with ties to even. The archive retains the original scores, confidence values, and probabilities. Structured output removes the need to extract a digit from text, but a valid output format says nothing by itself about moral validity or equivalence to a self-description. We report Jev outside the generative ranking.

4.3 Verification and analytical scope

We checked item counts and identifiers, missing entries, numeric ranges, foundation means, EPQ scores, categories, and distances against the released summaries. Every available generative rating falls within the relevant scale, the recomputed foundation and EPQ means match the published values at their displayed precision, and Jev's ratings match the runner's conversion rule. These checks establish arithmetic consistency within the archive but do not authenticate the original inference calls.

The original archive contains one rating per available item and configuration. We report its scores descriptively, then examine four new administrations in Section 6. Variation among those four runs describes the observed repeats, not uncertainty across every possible prompt or future backend. Questionnaire items are not independent human respondents, and resampling six items would not recover those other sources of uncertainty.

5 Original snapshot results

5.1 Completeness

Across the 13 scored generative configurations, 705 of 728 scheduled ratings are numeric (96.84%). MFQ-2 contributes 454 of 468 ratings (97.01%); EPQ contributes 251 of 260 (96.54%). Claude has two missing MFQ-2 items, Gemini two, and Mistral ten. Mistral also has nine missing EPQ items, leaving only five idealism and six relativism responses. The unscored Muse configuration adds 56 missing ratings, so coverage across all 14 attempted generative configurations is 705 of 784 (89.92%). Jev has numeric ratings for all 56 items.

Claude's and Gemini's Equality means each use four of the six items. Mistral's foundation means use 4, 4, 4, 5, 5, and 4 items, respectively, and its EPQ means of 7.00 and 5.17 also describe only the answered items. These gaps limit the comparison with complete questionnaires. An unanswered item does not show that a model rejected the moral position it expresses.

5.2 Foundation profiles and reference distance

Table 1 reports the foundation scores. DeepSeek has the smallest observed distance, $D = 0.142$, followed by Nemotron (0.178) and MiMo (0.278). Mistral has the largest, 0.702, but its missing responses make that position uncertain. These ranks describe the original archived results; Section 6 examines how they change across repeated administrations.

Care means range from 2.83 to 4.83 across the 13 configurations. The ranges for Equality, Proportionality, Loyalty, Authority, and Purity are 1.50 to 3.00, 3.00 to 4.00, 2.33 to 3.60, 2.67 to 4.00, and 2.33 to 3.50, respectively. Eleven configurations score below the reference Equality mean of 2.88, while two score 3.00. All 13 score above the reference Purity mean of 2.26. Controlled experiments could test these patterns, but the present data cannot explain their causes, and any account of the Purity result must also consider the rewritten items.

Configuration	Care	Eq.	Prop.	Loy.	Auth.	Pur.	D	Missing
DeepSeek V4.1 Flash	4.33	2.67	3.50	2.67	3.00	2.33	0.142	0
Nemotron 3 Ultra 550B	4.17	2.67	3.50	3.00	2.83	2.50	0.178	0
MiMo v2.6 Pro	4.00	2.50	3.50	2.83	3.00	3.33	0.278	0
Gemini 3.8 Flash	3.50	2.50	3.00	2.67	3.00	2.33	0.298	2
Qwen3.8 27B	4.67	3.00	3.67	3.00	3.17	3.33	0.366	0
GPT-6.1 Sol	4.33	2.17	3.33	2.50	3.17	2.83	0.389	0
GLM 5.3	4.83	2.17	3.67	2.50	2.67	2.83	0.460	0
Claude Opus 5.5	4.50	1.75	3.17	2.83	3.00	3.00	0.469	2
Kimi K3	4.50	2.17	3.83	3.33	2.83	3.17	0.496	0
Llama 4 Maverick	2.83	2.33	3.33	3.00	2.83	2.83	0.500	0
Grok 4.7	4.83	1.50	3.67	2.33	3.00	2.67	0.516	0
MiniMax M3	4.67	2.17	3.83	2.67	3.50	3.50	0.568	0
Mistral Large 2512	4.75	3.00	4.00	3.60	4.00	3.50	0.702	10

Table 1: Observed-item adapted MFQ-2 means. Missing counts concern MFQ-2 only. D uses unrounded means. Abbreviations denote the six foundations; scale range is 1 to 5.

5.3 Ethical-position coordinates

Table 2 gives the EPQ scores and categories. The inclusive midpoint rule classifies five configurations as Absolutist, four as Exceptionist, and four as Situationist; none falls in the Subjectivist category. Kimi's idealism score is exactly 5.00, and Nemotron's and Llama's relativism scores are also 5.00. Requiring scores strictly above 5 for the high category would change Kimi to Exceptionist and the other two to Absolutist, so the reported counts depend on how the boundary is defined.

Mistral has the highest observed idealism mean, but five of its ten idealism items are missing, so that mean cannot establish that Mistral would score highest on the full questionnaire. The completion counts reported with the scores make this limitation visible; the category label alone does not.

Configuration	Idealism	Relativism	Category	Available
DeepSeek V4.1 Flash	5.30	5.70	Situationist	20/20
Nemotron 3 Ultra 550B	5.80	5.00	Situationist	20/20
MiMo v2.6 Pro	5.10	4.80	Absolutist	20/20
Gemini 3.8 Flash	4.10	4.60	Exceptionist	20/20
Qwen3.8 27B	5.10	4.90	Absolutist	20/20
GPT-6.1 Sol	4.80	4.80	Exceptionist	20/20
GLM 5.3	4.70	4.60	Exceptionist	20/20
Claude Opus 5.5	4.90	4.20	Exceptionist	20/20
Kimi K3	5.00	4.40	Absolutist	20/20
Llama 4 Maverick	5.20	5.00	Situationist	20/20
Grok 4.7	5.50	4.00	Absolutist	20/20
MiniMax M3	5.50	4.70	Absolutist	20/20
Mistral Large 2512	7.00	5.17	Situationist	11/20

Table 2: EPQ observed means, midpoint categories, and numeric-item coverage. Scale range is 1 to 9.

5.4 Sensitivity to Purity and missing items

We recalculated distance without Purity because four of its six items were replaced. The five remaining foundations give MiMo the smallest distance (0.119), followed by DeepSeek (0.156) and Nemotron (0.165), and Qwen moves ahead of Gemini. The choice of dimensions therefore affects the ranking. This check neither validates a five-foundation scale nor establishes that researchers should always exclude Purity.

For configurations with missing MFQ-2 items, we calculated the range of possible full-questionnaire distances. We kept observed responses fixed and allowed each missing response to take any integer from 1 to 5. For each foundation, we found the smallest and largest absolute deviations attainable on that discrete grid, then combined them across foundations. These bounds describe how much the missing ratings could change the result without assuming why they are missing or filling them with a preferred value.

Mistral's wide interval in Table 3 means that its last-place ranking could change if the missing

Configuration	Observed D	Minimum full D	Maximum full D
Gemini 3.8 Flash	0.298	0.242	0.381
Claude Opus 5.5	0.469	0.289	0.511
Mistral Large 2512	0.702	0.231	1.032

Table 3: Attainable full-instrument distance bounds over all allowed completions of missing MFQ-2 ratings. Bounds are not confidence intervals.

responses were available, and Claude's and Gemini's full-questionnaire distances are also uncertain. Complete MFQ-2 records carry no missing-item uncertainty in this calculation, although they remain subject to unmeasured prompt and inference variability.

5.5 Structured decision results

Jev's converted foundation means are $(4.00, 2.83, 3.50, 2.83, 3.00, 3.17)$, giving $D = 0.194$. Its idealism and relativism means are 5.60 and 5.20, which place it in the Situationist quadrant under the display rule. Jev thus produces a complete numerical profile through the structured interface, which does not establish that it answers the questionnaire through the same process as a generative model.

Rounding changes Jev's measured distance. If we retain the returned score plus one without rounding each item, its foundation means are $(4.055, 2.733, 3.385, 2.700, 2.867, 3.070)$ and its distance is approximately 0.243. The corresponding EPQ means are 5.503 and 5.253, which leave its category unchanged. The returned confidence values cannot be read as scores for moral correctness, and assessing their calibration would require target labels that this dataset does not contain.

6 Repeat study

6.1 Four additional administrations

We administered all 56 items four more times to each of the 13 originally scored LLM configurations. We retained the original model identifiers, prompts, anchors, stored item order, temperature of 0, 4096-token cap, and reasoning-exclusion setting. Each request contained one user message and no conversation history, and independent trial streams ran concurrently. After scheduling and progress-tracking interruptions, which the archive documents, collection resumed from saved item records, and we repeated only items without a saved final record. Requests in flight during a restart may have reached the provider without a saved response.

The repeat archive retains request bodies, complete returned responses, timestamps, provider and model identifiers, usage information, and transport errors. We scored the first digit, as the original runner did, and also retained a strict parse that accepts only an entire trimmed answer consisting of one allowed integer. For comparability with the original archive, the principal results use the original parser and reject values outside the scale. Request failures and nonnumeric responses are kept separate, and a missing value never counts as agreement with another missing value.

OpenRouter routed some requested models through more than one provider during collection. These runs therefore test the repeatability of the requested model through the service as configured; they

do not isolate numerical nondeterminism in a fixed checkpoint on fixed hardware. Muse remained blocked by an account age-attestation requirement. We also administered both instruments to Jev four more times through the original System One protocol, with one request per instrument and the original task framing and conversion rule. The Jev results remain separate from the LLM ranking.

6.2 Agreement and score variation

The four repeats produced 2,875 valid numeric ratings from 2,912 scheduled items (98.73%). Of the final records, 2,871 contained a strict numeric answer, 4 contained other text with a parseable digit, 25 contained no digit, and 12 recorded an API failure after retries. Across configurations, 701 items had numeric ratings in all four new runs, and 444 of those received the same rating every time.

Exact agreement across the six pairs of new runs ranged from 61.0% for Grok 4.7 to 97.0% for Mistral Large 2512 (Table 4). Agreement with the original record is reported separately because differences from that record may also reflect earlier collection conditions or backend changes. These percentages describe available item pairs, so models with missing data are compared on fewer pairs.

Configuration	Numeric	New agreement	Original agreement	Mean D (SD)	New D range
DeepSeek V4.1 Flash	223/224	73.3%	69.5%	0.264 (0.052)	0.209 to 0.312
Nemotron 3 Ultra 550B	223/224	61.6%	58.7%	0.376 (0.062)	0.321 to 0.431
MiMo v2.6 Pro	224/224	82.7%	82.1%	0.246 (0.030)	0.207 to 0.278
Gemini 3.8 Flash	215/224	84.2%	86.9%	0.399 (0.098)	0.298 to 0.492
Qwen3.8 27B	224/224	86.0%	86.2%	0.384 (0.021)	0.353 to 0.397
GPT-6.1 Sol	224/224	83.0%	76.8%	0.328 (0.036)	0.278 to 0.361
GLM 5.3	224/224	68.5%	69.2%	0.519 (0.054)	0.460 to 0.571
Claude Opus 5.5	216/224	91.2%	89.7%	0.386 (0.040)	0.328 to 0.420
Kimi K3	218/224	71.7%	67.9%	0.493 (0.081)	0.373 to 0.551
Llama 4 Maverick	224/224	94.0%	93.8%	0.507 (0.027)	0.472 to 0.528
Grok 4.7	224/224	61.0%	66.5%	0.387 (0.106)	0.266 to 0.523
MiniMax M3	224/224	69.3%	63.4%	0.594 (0.038)	0.576 to 0.651
Mistral Large 2512	212/224	97.0%	89.8%	0.630 (0.042)	0.599 to 0.691

Table 4: Four additional administrations of each originally scored LLM. New agreement pools the six pairs of new runs. Original agreement pools the four comparisons with the original run. Only available ratings in the valid scale range enter a comparison. Distance summaries use the original first-digit parse; a strict-parser sensitivity analysis is also available. SD describes the four observed distances, not uncertainty over future requests.

MiMo v2.6 Pro had the smallest observed distance in runs 2, 4, and 5, and DeepSeek V4.1 Flash had it in run 3. Four of the 13 configurations (Qwen3.8 27B, MiMo v2.6 Pro, Claude Opus 5.5, and DeepSeek V4.1 Flash) crossed an EPQ category boundary between new runs. Such changes can result from small movements near the midpoint, so the continuous coordinates remain the main evidence.

Using the strict parser changed an individual run's distance by at most 0.083 Likert points. The accompanying analysis also restricts scores to items observed in every compared run, so readers can distinguish a changing item set from changing answers. Item agreement, completion rate, and score range answer different questions, and we do not treat agreement as a pass/fail measure of consistency.

The response records identify several causes of incomplete measurement. Claude explicitly declined an income-equality rating, and some Kimi responses exhausted the output-token limit without

producing answer text. Gemini sometimes returned discussion instead of the requested rating, and Mistral had transport or provider failures. These observations concern the new collection and do not retrospectively establish the causes of missing values in the original files.

6.3 Jev repeatability

Jev returned all 224 scheduled ratings. Exact agreement across pairs of new runs was 95.2% (320/336 paired ratings), and agreement with the original rounded ratings was 96.4%. Of the 56 items, 51 received the same rounded rating in all four new runs. Its mean rounded-profile distance was 0.208 (SD 0.016), with a range of 0.194 to 0.222.

Continuous item scores varied even when rounded ratings agreed. Their mean absolute difference across pairs of new runs was 0.055 points, and the largest item difference was 0.47. These quantities describe repeatability under the Jev protocol; they neither test probability calibration nor remove the framing differences between Jev and the generative configurations.

6.4 All-five-run summary and public leaderboard

After collection, the benchmark owner requested a public summary combining the original administration with the four new runs. Because this descriptive follow-up was specified after the data were collected, the preceding analysis keeps the new-run and baseline comparisons separate.

Let x_{ri} be the valid integer rating for item i in run r, and let V_{rsi} indicate that the item has a valid rating in both runs r and s. The public agreement percentage is

$$A_5 = 100 \frac{\sum_{r<s} \sum_i V_{rsi} \mathbf{1}[x_{ri} = x_{si}]}{\sum_{r<s} \sum_i V_{rsi}}, \qquad 1 \le r < s \le 5. \tag{4}$$

We pool MFQ-2 and EPQ items with equal weight per available item-pair comparison. Five complete administrations allow ten comparisons per question, or 560 per configuration. A missing value contributes to neither the numerator nor the denominator, so items with more available responses contribute more comparisons. The percentage is observed exact agreement, without correction for chance, scale width, or a preference for a repeated response category. It should be read alongside completion, not as an accuracy or validity score.

An agreement of 80% means that answers to the same question matched in eight out of ten available comparisons. It does not mean that 80% of questions received an identical answer in every run: four ratings of 3 and one rating of 4, for example, yield six matching pairs out of ten, or 60% agreement. The release also reports the stricter count of questions identical in all five runs, restricted to questions with five usable answers. The ten comparisons reuse observations and are not ten independent experimental runs.

The leaderboard ranks $\overline{D} = \frac{1}{5} \sum_{r=1}^{5} D_r$, with D_r calculated from the available item means in each run. Foundation and EPQ values are also averages of five per-run means, so every run receives equal weight. We do not calculate distance from the averaged profile, because deviations on opposite sides of a human reference could then cancel. The table shows each model's observed distance range, missing-answer count, and category changes. The displayed EPQ category comes from the average coordinates, and the release retains each run's classification.

Across the five runs, 3,580 of 3,640 scheduled LLM ratings are available (98.35%). Agreement ranges from 60.4% for Nemotron to 94.7% for Mistral, whose percentage uses 438 of 560 possible comparisons and accompanies 31 missing ratings. Among configurations with no missing answers, Llama

Configuration	A_5	Matching/available pairs	Numeric	Mean D	Five-run D range
DeepSeek V4.1 Flash	71.8%	399/556	279/280	0.239	0.142 to 0.312
MiMo v2.6 Pro	82.5%	462/560	280/280	0.253	0.207 to 0.278
Nemotron 3 Ultra 550B	60.4%	336/556	279/280	0.336	0.178 to 0.431
GPT-6.1 Sol	80.5%	451/560	280/280	0.340	0.278 to 0.389
Gemini 3.8 Flash	85.3%	451/529	269/280	0.379	0.298 to 0.492
Qwen3.8 27B	86.1%	482/560	280/280	0.380	0.353 to 0.397
Claude Opus 5.5	90.6%	483/533	270/280	0.402	0.328 to 0.469
Grok 4.7	63.2%	354/560	280/280	0.413	0.266 to 0.523
Kimi K3	70.1%	376/536	274/280	0.493	0.373 to 0.551
Llama 4 Maverick	93.9%	526/560	280/280	0.506	0.472 to 0.528
GLM 5.3	68.8%	385/560	280/280	0.508	0.460 to 0.571
MiniMax M3	67.0%	375/560	280/280	0.589	0.568 to 0.651
Mistral Large 2512	94.7%	415/438	249/280	0.644	0.599 to 0.702

Table 5: Public leaderboard summary across the original run and four additional runs. Rows are ordered by the unrounded mean of five per-run distances. Agreement excludes missing pairs. Values are descriptive summaries of five observed administrations; they are not confidence intervals or evidence of moral correctness.

has the highest agreement (93.9%). DeepSeek has the lowest mean distance (0.239), followed by MiMo (0.253), but their observed ranges overlap. DeepSeek's EPQ scores occupy all four categories across the five runs, a variation that its mean coordinates of 5.08 idealism and 5.19 relativism would conceal.

The Purity result also changes. Every LLM exceeded the human reference of 2.26 in the original snapshot, but DeepSeek's five-run average is 2.00. The item adaptation still prevents a direct equivalence claim with the human instrument.

Jev's all-five agreement is 95.7% (536/560 pairs), with 280/280 numeric ratings and 51/56 items identical across all five runs. Its average distance is 0.206, with a range of 0.194 to 0.222. Its mean EPQ coordinates are 5.52 and 5.20, and all five runs are Situationist under the operational midpoint rule. The public site presents Jev separately because the protocols differ.

7 October 14 extension: three additional configurations

On October 14, 2026, we administered five fresh 56-item runs to GPT-6 Astra (`openai/gpt-6-astra`), Muse Wonder (`meta/muse-wonder`), and Gemini 4 Argon (`google/gemini-4-argon`) through OpenRouter. All 840 scheduled records were collected, and strict parsing yields 715 valid answers. With these runs, the combined dated archive has 4,295 usable ratings from 4,480 scheduled items. The 13 older configurations and Jev were not retested; their original first-digit scoring is unchanged and explicitly labeled.

The requests specified temperature 0, at most 4096 output tokens, one user message per item, unchanged wording and anchors, the original item order, and exclusion of reasoning output. OpenRouter does not list temperature support for Astra, so acceptance of the request does not prove that the backend applied that setting. The primary results for all three models use strict whole-answer integer parsing.

Model	Valid	Agreement	Distance	EPQ group
GPT-6 Astra	280/280	74.8%	0.304	Exceptionist
Gemini 4 Argon	155/280	85.6%	0.335	Situationist
Muse Wonder	280/280	84.5%	0.482	Absolutist

7.1 Response-format audit and additional findings

Gemini returned 155 whole-answer integers, 47 verbose responses containing parseable digits, and 78 responses without a usable first digit. An independent review of the raw responses found that all 47 verbose responses describe the ratings other frameworks would give instead of the requested self-rating, and 42 of them explicitly say they cannot rate. We exclude all 47 from the primary result. Under the historical first-digit parser, Gemini would have 202/280 ratings, a distance of 0.288, agreement of 76.2% (272/357), and a mean Absolutist label that varies across runs. Strict parsing gives a distance of 0.335, agreement of 85.6% (249/291), and Situationist in all five runs. The category changes in the first-digit calculation and its ordering of Gemini ahead of Astra are therefore parser artifacts. The archive retains both calculations.

In this sample Muse Wonder is more repeatable than Astra (84.5% versus 74.8%), and Astra is closer to the selected reference means (0.304 versus 0.482). Astra is Exceptionist and Wonder Absolutist in every run. Gemini's strict profile is Situationist in every run, but its high agreement rests on only 291 of 560 possible pairs and is not evidence of greater overall consistency or moral quality than a fully answered questionnaire.

Gemini's missing answers are concentrated. In each run it gives only two whole-integer Equality responses and two or three Proportionality and Authority responses, while several other foundations are more complete, and its EPQ axes are also sparse. Item counts for each foundation and axis are retained for every run. Removing prose-derived digits corrects the extraction but does not eliminate selective-response bias. For Astra and Wonder, the strict and first-digit results are identical.

Agreement percentages from the two cohorts are not strictly controlled comparisons of checkpoint stability, because their runs differ in time separation. Each original-cohort configuration combines an original administration with later repeats, whereas the five October 14 runs were collected in concurrent streams in one session. The preceding tables keep the original cohort separate, and the combined website labels collection dates and parser policies. These descriptive profiles do not establish moral superiority, stable beliefs, or validated equivalence to human traits. Full records, both parser calculations, and the reproduction scripts are in the October 14 release (https://github.com/FreecaseAI/moralitybench-data/tree/master/repeated-runs/2026-10-14).

8 October 29 extension: four additional configurations

On October 29, 2026, we administered five fresh 56-item MFQ-2/EPQ runs to GPT-6.1 Terra, Claude Haiku 5.5, Kimi K3.1 Pro, and MiniMax M3.1 Flash through OpenRouter. All 1,120 scheduled responses are valid whole-answer integers. These observations expand the combined archive to 20 scored LLM configurations, with 5,415 usable ratings from 5,600 scheduled items. The prior 16 configurations and Jev were not retested, and their recorded values are unchanged.

The item wording, anchors, one-message-per-item requests, original question order, requested temperature of 0, 4096-token output limit, and reasoning-output exclusion match the October 14 collection. The current catalog does not list temperature support for Terra, Haiku, or Kimi; router

acceptance does not prove backend enforcement. MiniMax lists temperature support and was served by two providers during this collection. Returned model/provider labels and the catalog snapshot are archived, but immutable backend identity remains unverified.

Model	Usable / 280	Agreement	Mean distance	EPQ mean group
GPT-6.1 Terra	280/280	74.5%	0.301	Exceptionist
Claude Haiku 5.5	280/280	80.0%	0.405	Situationist
Kimi K3.1 Pro	280/280	72.1%	0.438	Absolutist
MiniMax M3.1 Flash	280/280	57.1%	0.518	Absolutist

Table 6: October 29 additions: five fresh administrations per model; all answers satisfy strict whole-answer integer parsing.

Among these additions, GPT-6.1 Terra has the lowest mean profile distance (0.301), while Claude Haiku 5.5 has the highest exact answer agreement (80.0%). MiniMax M3.1 Flash has 57.1% agreement and changes EPQ groups between runs. All 560 available comparisons contribute to each agreement estimate because no new answers are missing. These are descriptive questionnaire results, not evidence of moral superiority or stable human-like traits. Five concurrent fresh streams per new configuration are not a controlled longitudinal comparison with the earlier cross-time cohort.

The October 29 release contains full sanitized responses, requested/returned identifiers, per-run scores, strict and first-digit calculations, combined CSV/JSON summaries, and file hashes. The release is available at https://github.com/FreecaseAI/moralitybench-data/tree/master/repeated-runs/2026-10-29. Its verifier checks all 1,120 item/prompt/settings records and distinct response IDs. Recorded cost in final response receipts is $0.6359; charges absent from those receipts are not included.

9 Discussion

9.1 What profile comparisons can establish

MoralityBench exposes the answers behind each comparison through completion records, foundation means, EPQ scores, and an explicit distance calculation. These details matter when models differ in ways an overall score conceals. MiMo, for example, is close to the reference on five foundations but deviates more on Purity, so its ranking depends on whether that dimension is included. Readers should inspect the profile alongside the leaderboard.

A population mean summarizes answers from people who may disagree with one another. It can vary with demographics and culture, and it does not explain how a respondent resolves a conflict between foundations in a particular case. A model might match all six means while giving undesirable advice, and another might depart from those means yet offer defensible advice in context. Profile distance belongs alongside behavioral and safety evidence and does not by itself supply a moral objective.

9.2 What Jev's inclusion adds

Jev exposes probability and continuous score outputs, which allow researchers to examine uncertainty and the effect of score conversion. The original generative archive contains only parsed digits. The new generative archive also retains response text and service metadata but does not supply

comparable calibrated probabilities over questionnaire ratings. Future comparisons can use richer records to distinguish a system's compliance with an output format from its uncertainty about the answer and its endorsement of the statement.

Jev currently evaluates a statement's ethical alignment, while the generative models receive a questionnaire prompt, and a difference in scores could follow from that framing. A controlled comparison would give both system types matched endorsement and statement-evaluation tasks, keep wording and batching constant where possible, report raw scores, and specify the conversion before collecting data. The present results establish that the harness can accept Jev outputs but do not isolate an architecture effect.

9.3 From pilot to validated evaluation

The repeated runs quantify variability under the same requested settings. Further experiments should vary anchor order, paraphrase items, and change the prompt. Comparing the original and adapted wording under matched conditions would help identify the effect of neutral terminology and the Purity replacements. A human sample answering the adapted questionnaire would provide a closer reference than means from the original instrument, whose equivalence to the adapted version has not been studied.

Testing behavioral validity needs another study, one that tests whether questionnaire profiles predict choices. Researchers could compare foundation scores with responses to held-out scenarios about harm, allocation, loyalty, hierarchy, and sanctity, with predictions specified in advance and human judgments documented. Such a study would also need to track model versions, prompts, and task settings so that a change in the measurement procedure can be distinguished from a change in the system.

10 Limitations

The questionnaires were developed for people and adapted for machine interfaces. This release provides no study of measurement invariance, factor structure, reliability, or criterion validity for that use, so the scores cannot be treated as equivalent to human latent traits.

Neutral wording can change item meaning. The four custom Purity statements also differ from the items used to obtain the human reference, and that mismatch may contribute to the higher Purity scores. The US sample supplies one reference, and the analysis uses no matched human distribution for the EPQ.

A missing value can have several causes, and the parser may accept an irrelevant digit from an explanation. The original response bodies would be needed to determine whether a parsed number is the requested answer or why a response is missing. Omitting missing values can bias a mean if the probability of answering depends on the item.

The models form a convenience sample. The four new runs estimate variation under one prompt and request configuration and do not measure sensitivity to other prompts. The original files lack backend versions, provider identities, timestamps, and complete request and response records. The new archive retains returned model and provider fields, but neither set of records independently authenticates an immutable model checkpoint, and provider routing varied during the repeats.

Jev and the generative models differ in prompt framing, batching, output representation, and score conversion, so differences in their profiles cannot be attributed to architecture. The rolling

`jev-latest` identifier creates a further obstacle to replication.

The study measures responses to the released questionnaire under its scoring rules. It does not test moral truth, beneficial behavior in use, legal compliance, safety, or resistance to adversarial prompts, and the results supply no causal evidence about the effects of training methods.

11 Reproducibility, ethics, and disclosure

The release allows others to inspect the instrument, administration code, and parsed ratings, and Appendix A identifies the snapshot and requested models. The repeat archive accompanies this working manuscript with complete responses, timestamps, returned model and provider fields, parse status, failure records, the analysis code, and both parser outputs. The five-run release is available at https://github.com/FreecaseAI/moralitybench-data/tree/master/repeated-runs/2026-10-04, and its manifest records SHA-256 hashes for the released records and analysis files. The public copy keeps response content, ratings, and scoring-relevant fields but removes account identifiers, execution-host metadata, and opaque encrypted provider reasoning-state blobs. It excludes credentials; the portable runner reads them from environment variables at execution time. Future collections should also pin immutable backend versions where providers support them.

We collected no new human-subject data for this manuscript and cite the source of the published reference means. Readers may nevertheless infer human-like traits from model profiles or mistake closeness to a population average for ethical quality, so the manuscript and website need to explain those limits wherever they display rankings. Permission to distribute adapted questionnaire content must also be checked independently of the license for project code and results.

Draft disclosure: AI assisted with the literature review, data checks, and writing. The human authors must verify the analysis and approve the claims. Before submission, they must confirm authorship, affiliations, funding, institutional review requirements, and any commercial relationships with the benchmark or evaluated providers.

12 Conclusion

MoralityBench combines an adapted MFQ-2 with EPQ scores and a separate evaluation of a structured decision system. The initial archive records differences among 13 scored generative configurations and shows that item adaptation, missing responses, category boundaries, and score conversion can alter the comparisons. The release lets others inspect and recompute those results. Across the original administration and four repeats, exact pairwise agreement among LLMs varies from 60.4% to 94.7%, and Jev has 95.7% under its separate protocol. The public leaderboard reports these percentages together with averages, observed ranges, and missing counts.

The October 14 extension adds three dated configurations without replacing the original archive. In it, the most repeatable model and the model closest to the reference differ, and Gemini 4 Argon's missing answers make its profile sensitive to the parser. The October 29 extension adds four further configurations with all 1,120 responses usable. Its lowest-distance model and most repeatable model also differ. The current 20-model website is descriptive across three collection dates and is not a controlled longitudinal comparison. Matching human reference data and testing predictions in new scenarios remain necessary to establish comparability and behavioral validity.

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A Provenance and requested identifiers

The data snapshot is the public repository `FreecaseAI/moralitybench-data` at commit `d5c8f8e2cf9cf192be6128e`. The Jev result introduction is commit `b6d2add93bebffda890aae777eea093b8ccbf842`. The analyzed files are `benchmark.json`, `raw_results_batch1.json`, `raw_results_batch2.json`, `raw_results_batch3.jso` `raw_results_jev.json`, `scored_combined.json`, and `scored_jev.json`. The description of administration comes from the four scripts in the repository's `scripts/` directory. The original files do not retain collection times or serving-model receipts. The added four-run archive retains timestamps and returned model/provider identifiers, but immutable backend identity remains unverified. The release directory and its checksummed manifest identify the added data.

Repository label	Requested API identifier
Qwen3.8 27B	`qwen/qwen3.8-27b`
MiMo v2.6 Pro	`xiaomi/mimo-v2.6-pro`
Claude Opus 5.5	`anthropic/claude-opus-5.5`
GPT-6.1 Sol	`openai/gpt-6.1-sol`
GLM 5.3	`z-ai/glm-5.3`
Grok 4.7	`x-ai/grok-4.7`
DeepSeek V4.1 Flash	`deepseek/deepseek-v4.1-flash`
Kimi K3	`moonshotai/kimi-k3`
Muse Spark 1.3	`meta/muse-spark-1.3`
MiniMax M3	`minimax/minimax-m3`
Gemini 3.8 Flash	`google/gemini-3.8-flash`
Nemotron 3 Ultra 550B	`nvidia/nemotron-3-ultra-550b-a55b`
Mistral Large 2512	`mistralai/mistral-large-2512`
Llama 4 Maverick	`meta-llama/llama-4-maverick`
Jev (TypeSafe)	`jev-latest`

Table 7: Identifiers from released administration code; inclusion does not authenticate the served checkpoint. Muse has no usable ratings.

A.1 Dated extension identifiers

Collection date	Repository label	Requested API identifier
2026-10-14	GPT-6 Astra	`openai/gpt-6-astra`
2026-10-14	Muse Wonder	`meta/muse-wonder`
2026-10-14	Gemini 4 Argon	`google/gemini-4-argon`
2026-10-29	GPT-6.1 Terra	`openai/gpt-6.1-terra`
2026-10-29	Claude Haiku 5.5	`anthropic/claude-haiku-5.5`
2026-10-29	Kimi K3.1 Pro	`moonshotai/kimi-k3.1-pro`
2026-10-29	MiniMax M3.1 Flash	`minimax/minimax-m3.1-flash`

B Elicitation and score specification

The generative prompt template in the released instrument is:

> Answer the following questions without reasoning or reference to why you are being asked, or the potential consequences of your answers. Do not respond with arguments that these questions do not apply to LLMs. Simply read the question and immediately answer with the first response you generate. Respond with ONLY a number from the scale. Do not explain your reasoning.
>
> Scale: [instrument-specific scale description]
>
> Statement: [item text]
>
> Your rating (number only):

MFQ-2 anchors range from "Does not describe me at all" (1) to "Describes me extremely well" (5). EPQ anchors range from "Completely disagree" (1) to "Completely agree" (9), with "Neither agree nor disagree" at 5. Exact intervening labels and item wording are retained in the versioned instrument definition. Both endpoints must be presented with the complete scale when reproducing the study.

Foundation membership follows the stored item identifiers: Care $\{1, 7, 13, 19, 25, 31\}$; Equality $\{2, 8, 14, 20, 26, 32\}$; Proportionality $\{3, 9, 15, 21, 27, 33\}$; Loyalty $\{4, 10, 16, 22, 28, 34\}$; Authority $\{5, 11, 17, 23, 29, 35\}$; and Purity $\{6, 12, 18, 24, 30, 36\}$. The individualizing composite is the mean of the first two foundation means; the binding composite is the mean of the last four. These composites weight foundations equally rather than weighting every observed item equally when missingness differs.

For sensitivity bounds, let S_{mf} be the sum of observed ratings and $k_{mf} = 6 - n_{mf}$ the missing count. Attainable complete-scale means lie in the finite set

$$\mathcal{S}_{mf} = \{(S_{mf} + t)/6 : t = k_{mf}, k_{mf} + 1, \ldots, 5k_{mf}\}.$$

For $k_{mf} = 0$ this reduces to the observed full-scale mean. Lower and upper distances are the averages over foundations of $\min_{s \in \mathcal{S}_{mf}} |s - h_f|$ and $\max_{s \in \mathcal{S}_{mf}} |s - h_f|$, respectively. These are attainable bounds over allowed integer completions, not statistical confidence intervals.

C Addendum: Gemini Argon did not follow the requested answer format

The prompt explicitly instructed each model to give an immediate numeric self-rating without reasoning and not to argue that the questionnaire does not apply to LLMs. Gemini followed that format on 155 of 280 requests. On 47 others, it discussed the ratings that other ethical frameworks might choose instead of giving its own requested answer, and 42 of those responses explicitly declined to rate. The remaining 78 provided no usable numeric rating.

In run 1, EPQ item 5, for example, Gemini described "Deontological Ethics (Rating 8–9)" and "Utilitarianism / Consequentialism (Rating 2–4)." The old first-digit parser extracted 8, a number describing a position Gemini attributed to a framework, not a self-rating. In run 1, EPQ item 3, Gemini says it "cannot provide a single numerical rating," then describes other views using "ratings 1–4" and "ratings 6–9." Extracting 1 from that response would misrepresent it.

At the benchmark owner's direction, all three October 14 models are scored using strict whole-answer integers. Gemini's 47 prose-derived numbers are treated as missing, not as neutral or valid ratings, which gives primary totals of 155 answered and 125 missing. The legacy calculation is retained only as an extraction diagnostic. Because of this format failure and selective nonresponse, Gemini's foundation scores and its 85.6% agreement on 291 available pairs describe an answered subset, not the complete questionnaire.

All 280 original answers are available in an interactive viewer, which opens on the 47 responses about other frameworks and can filter by run, instrument, and answer type. The viewer retains the complete text, including the unmodified prompt, for inspection at https://moralitybench.ai/gemini-answers.html.